ML04 Decisions and a hand-written flower rule

Week 2, Python you need | Session 4 of 24 | Machine Learning

Today's question. How often does a three-line rule name the right flower?

Objective. By the end: you can write an if, elif and else decision, test it at its boundary, and count how often a simple rule is right.

> **WHERE THIS IS USED**
> Apps decide with rules such as: if the balance is too low, refuse the payment. Botanists also tell flowers apart by measuring petals, and today you write a three-line rule that does it. Can your rule beat a guess?

Before this session: ML03 Arithmetic, converting text and reading errors. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
comparison	A question about two values that answers True or False.	Asking whether your money is at least the ticket price.
Boolean	A value that is either True or False.	A light switch that is on or off.
if	Runs the indented lines below it only when its question is True.	If it rains, take an umbrella.
else	Runs when the if question was False.	Otherwise, leave the umbrella.
elif	Asks a second question when the first was False.	If it is raining take an umbrella, otherwise if it is cloudy take a hat.
indentation	Four spaces at the start of a line that show which instructions belong together.	Writing a sub-point under a heading.

Block A. A decision with if and else

A student has UGX 10000 and an item costs UGX 8000. Run the cell, then change the cost.

Predict. Predict the message that will print. Write your guess on paper first.

Your guess, written before you run it

```
budget = 10000
cost = 8000
if cost <= budget:
    message = "Within budget"
else:
    message = "Over budget"
```

```
print(message)
```

Run it, then compare. The cell prints Within budget. cost <= budget is a comparison and its answer is True, which is a Boolean. if runs the indented lines when the answer is True and else runs them when it is False. The four spaces at the start of the lines are indentation.

Investigate.

1. What is the answer to 5000 >= 5000?
2. What does = do compared with ==?

Modify. Set cost to 12000 and run the decision again.

Block B. A rule that names a flower

The iris data records flowers of three species. Petal length helps tell them apart. Run the rule for one flower, then try the others on the table.

Predict. Look at the rule. Predict which species it names for a petal length of 4.7 centimetres. Write your guess on paper first.

Your guess, written before you run it

```
petal_length = 4.7
if petal_length < 2.5:
    guess = "setosa"
elif petal_length < 4.8:
    guess = "versicolor"
else:
    guess = "virginica"
print(guess)
```

Run it, then compare. The cell prints versicolor. elif asks a second question only when the first answer was False. Try the eight petal lengths. The true species are setosa twice, versicolor three times and virginica three times. The rule gets seven right and misses the 4.9 flower.

Investigate.

1. Which of the eight flowers does the rule get wrong?
2. How would you move the rule so that 4.9 is right?
3. What is the accuracy on these eight flowers?

Modify. Set petal_length to 4.9 and run the rule.

Make

A ticket costs UGX 5000. Write a decision that prints Buy when the budget is at least 5000 and Save otherwise. Test 4999, 5000 and 5001. Then say how many of the eight flowers your flower rule names correctly.

Save your work as ML04. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
IndentationError: expected an indented block	The line after if, elif or else does not start with spaces.	Add four spaces at the start of each line inside the block.
SyntaxError: invalid syntax	A colon is missing at the end of an if, elif or else line.	Put a colon at the end of the line.

Exit ticket

A decision runs different lines depending on a True or False question, and a simple rule can be tested by counting how often it is right.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.